Introduction to R

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Module 0 (optional)

Computational Methods in Statistics and Data Science (DATASCI 406)

What is R?

When we say, “R” we are referring to three interrelated things:

- A language
- A community
- An implementation or environment

R: The language

R is a **statistical programming language**:

- R is specifically design to load, manipulate, and analyze tabular data (versus Python, Java, C++)
- We can use R to easily code up new algorithms, methods (versus Stata, SAS)
- We interact with R via scripts containing textual input (versus Minitab, Excel)

Key concepts:

- Store data in variables, usually **vectors**, **matrices**, and **data frames**.
- Manipulate data using **functions**, **iteration**, and **high level declarations**.
- Process data using **scripts** and **RMarkdown documents**.

R: The community

The **Comprehensive R Archive Network** is a collection of **user submitted packages**, currently more than **20,000**.

R is supported via: textbooks, official mailing lists, StackOverflow, R Bloggers, YoutTube, etc (though it can be hard to Google for sometimes).

R is being adopted by Fortune 500 companies, government, start ups, applied academic disciplines, many others.

R: The environment

The official R implementation consists of an **command line interface** for entering R commands, a **batch file processor** for handling scripts, and a basic **graphical user interface** for handling plots.

We will be using **RStudio**, installed on your own machine, which adds:

- Projects to handle multiple R files, data files.
- More complete file editor with syntax completion
- Help system and graph/plot display
- Integration with external software development tools
- RMarkdown to HTML and PDF support

RMarkdown

RMarkdown is a **plain text file** that contains **structured text** and **embedded R code**. It can be **processed** into a PDF or HTML file. The R is **evaluated** and the **results added to the file**, including plots.

Some great features:

- Put the description and the implementation in one place.
- Inline R code allows printing out values – no more copy and paste errors.
- Easy to provide instructions and starter code for projects.
- Includes a math language for writing up analytical results.

DATASCI 406 and R

Why do we use R in DATASCI 406 (instead of another language):

- Algorithms written in R tend to be slower, but speed is not our primary goal, understanding is.
- R contains useful distributions that we can sample from when doing Monte Carlo
- Reference implementations to compare/use for many algorithms
- R's baked in **vectorization** allows for focusing on the math and stats instead of programming overhead
- Great data manipulation and visualization tools

Exercises